DO NOW — RECALL

Synoptic retrieval: data structures

Name: _____ Class: _____ Date: _____

1. A three-element array is accessed at index 3. What happens?

2. What is the worst-case Big O of a sequential search?

3. Push A, B, C. Which pops first?

4. In five buckets, h(k)=k MOD 5. Where does 12 start?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Index 3 is outside indexes 0–2; a C# IndexOutOfRangeException occurs.

2. Model answer

O(n), as it may inspect every element.

3. Model answer

C, because a stack is LIFO.

4. Model answer

Bucket 2.

Vocabulary — know these terms

LIFO — Last In, First Out.

FIFO — First In, First Out.

head — Reference to the first node in a linked list.

BST — Binary search tree ordered by key comparisons.

collision — Two hashed keys initially share a bucket.

NEW LEARNING

Read, annotate and explain

Retrieve operations

A stack pops its newest item; a queue dequeues its oldest item.

Trace changing state

After Push 3, Push 6, Pop, Push 9, the stack is [3,9]. After Enqueue A,B then Dequeue, the queue is [B].

Apply the BST rule

Insert 40 first, 20 left, 60 right, then 30 as the right child of 20.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

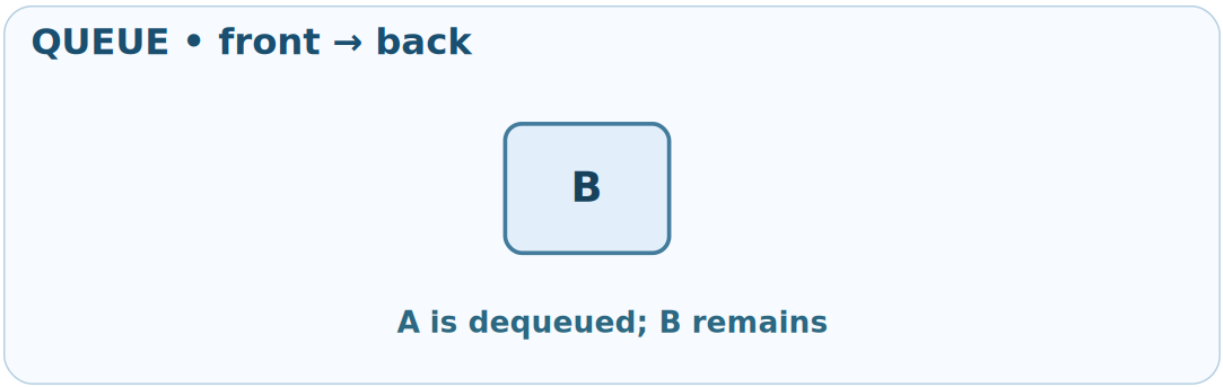
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



STACK • bottom → top 3, 9 Pop removes 6, then push 9

New Learning

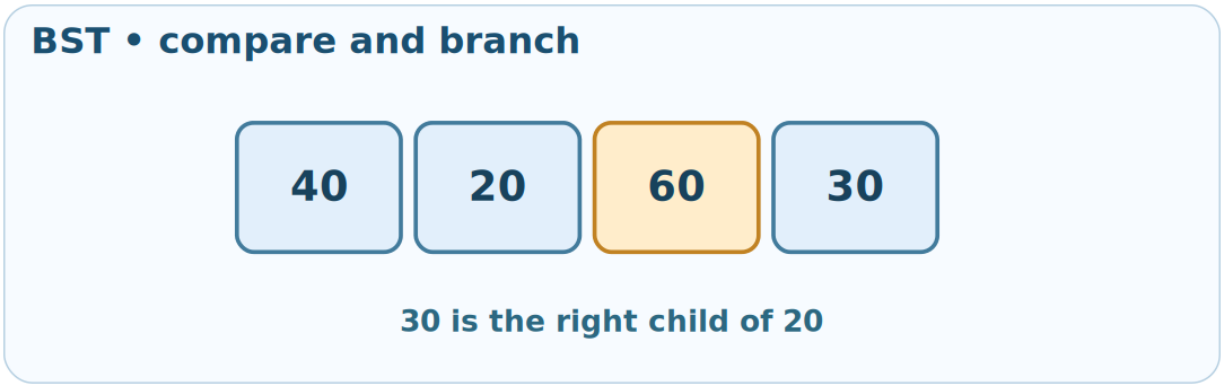


QUEUE • front → back B A is dequeued; B remains

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



BST • compare and branch 40, 20, 60, 30 30 is the right child of 20

I DO — WORKED EXAMPLE

Follow the worked method

Trace stack operations Push(2), Push(7), Pop(), Push(5).

Predict first

Start with [2,7]. The top of the stack is 7.

Check the method

Pop removes 7, leaving [2].

Check the method

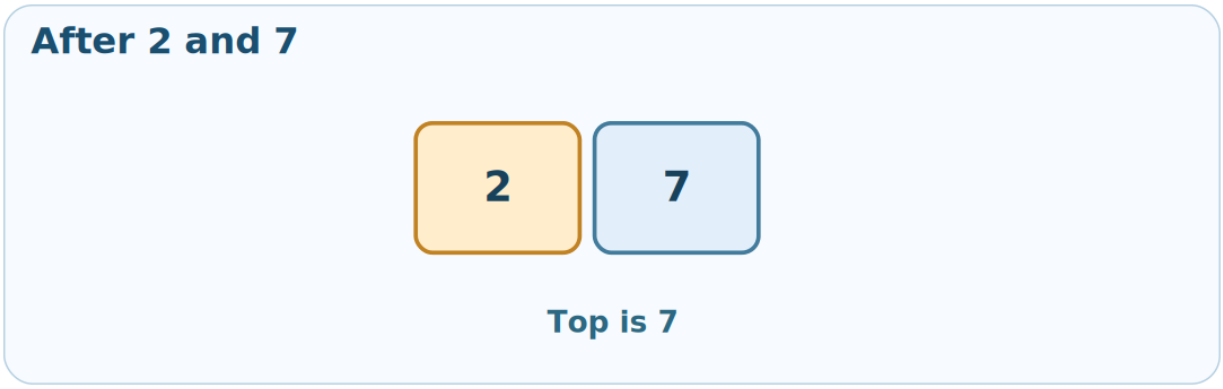
Push 5 gives [2,5]; the top is now 5.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace stack operations Push(2), Push(7), Pop(), Push(5).

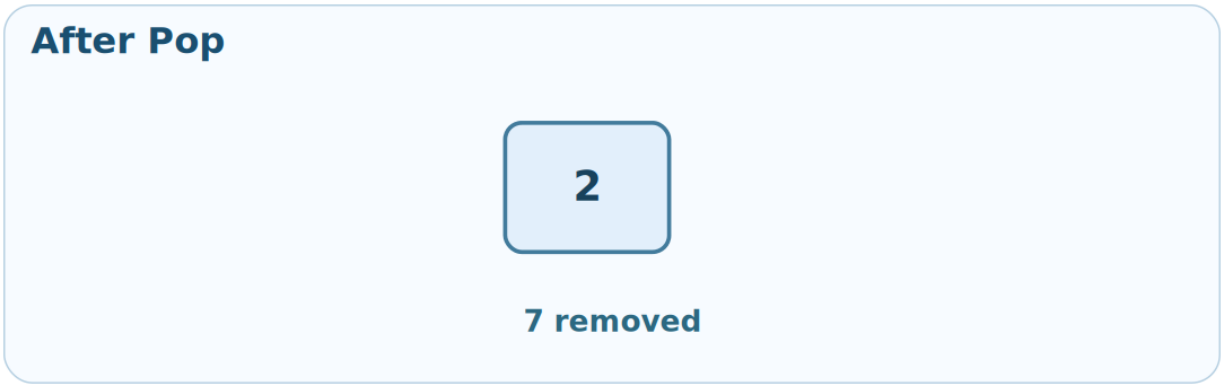
Start with [2,7]. The top of the stack is 7.



After 2 and 7 2, 7 Top is 7

Trace stack operations Push(2), Push(7), Pop(), Push(5).

Start with [2,7]. The top of the stack is 7.



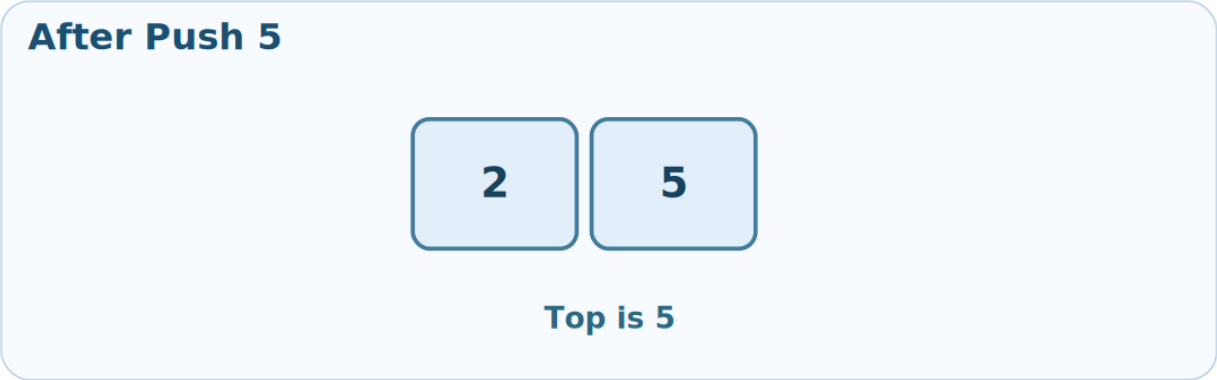
After Pop 2 7 removed

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace stack operations Push(2), Push(7), Pop(), Push(5).

Start with [2,7]. The top of the stack is 7.



After Push 5 2, 5 Top is 5

WE DO — GUIDED PRACTICE

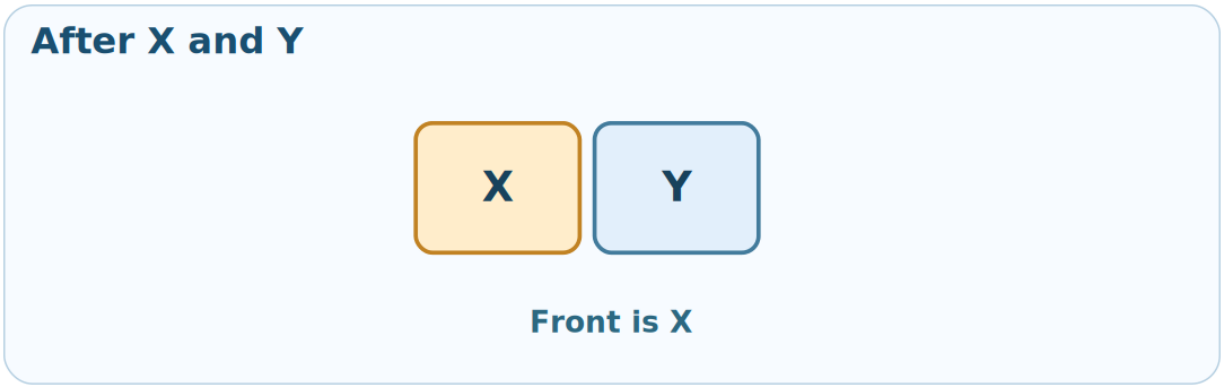
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace queue operations Enqueue(X), Enqueue(Y), Dequeue(), Enqueue(Z).

After the two enqueues, the queue is [X,Y].



After X and Y X, Y Front is X

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first After the two enqueues, the queue is [X,Y].

DISCUSS — Compare the sequence of steps or states. Identify the first difference in your methods and justify which step is correct.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

A stack contains [4,9] bottom-to-top. Push 2, Pop, Pop. State both removed values and remaining stack.

Question 2

A queue receives 5,8,3. It dequeues once then enqueues 2. Give the queue and explain why its next dequeue removes 8.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Insert 40,20,60,10,30,25 into a BST. Give the parent of 25, state whether it is a left/right child, and explain what happens to worst-case height when inputs arrive sorted.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A queue normally removes the most recently enqueued item.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

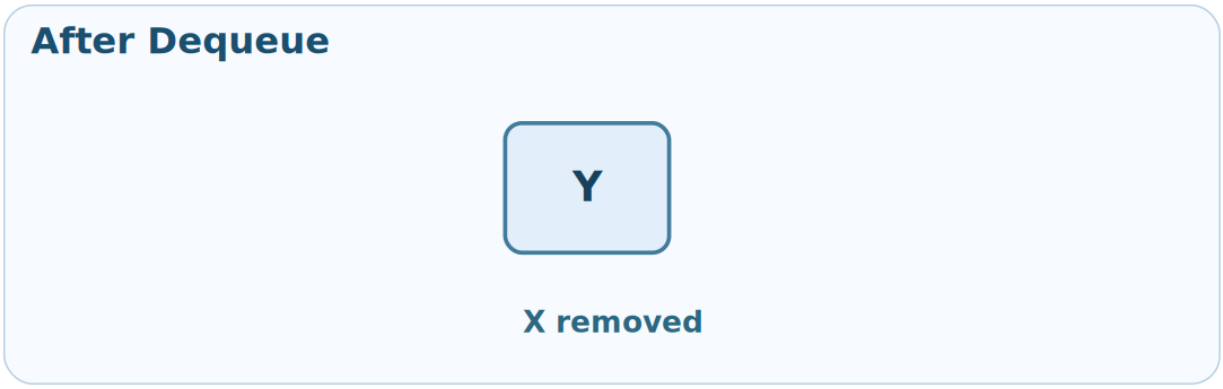
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace queue operations Enqueue(X), Enqueue(Y), Dequeue(), Enqueue(Z).

After the two enqueues, the queue is [X,Y].



After Dequeue Y X removed

Trace queue operations Enqueue(X), Enqueue(Y), Dequeue(), Enqueue(Z).

After the two enqueues, the queue is [X,Y].



After Enqueue Z Y, Z Next removed is Y

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Check the method

Dequeue removes X, leaving [Y].

Enqueue Z produces [Y,Z]. The next dequeue removes Y.

IMPROVE: Explain one correction to your own reasoning.